

# Shifted Catalan Hankel Polynomials

Exact coefficients, sharp recurrences,  
and a denominator conjecture

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## Abstract

For the Catalan numbers  $C_n$ , consider the size- $N$  determinant  $D_N^{(m)}(a, b) = \det(aC_{i+j+m} + bC_{i+j+m+1})_{0 \leq i, j < N}$ . We give a self-contained proof of the denominator-exponent conjecture stated in Paul Barry's 2020 note on this family. With  $d = \binom{m}{2}$ , the generating function is  $P_m(t)/(1 - (a + 2b)t + b^2t^2)^{d+1}$ . The denominator is minimal whenever  $b \neq 0$ ,  $a \neq 0$ , and  $a \neq -4b$ . We determine every exceptional case, prove that  $\deg P_m = (m - 1)^2$  for  $b \neq 0$ , and establish a reciprocal symmetry of the numerator. An explicit positive-integer coefficient formula also identifies and repairs a one-unit shift in the source's separate Conjecture 2. The essential degree reduction is a Vandermonde cancellation in confluent Christoffel minors. It yields an exact sum of two exponentials with polynomial coefficients, a finite full asymptotic expansion, and an explicit first correction. Independent integer and rational computations, including direct determinants and polynomial-identity certificates at fixed sizes, accompany the proofs.

**Main result.** For all nonexceptional complex parameters,

$$\text{minimal constant-coefficient recurrence order} = m(m - 1) + 2.$$

The proof is uniform in the determinant size and the Catalan shift.

**Status and scope.** This is an unrefereed, AI-assisted mathematical manuscript, not a formalized proof. Its theorems are supported by explicit arguments, not inferred from finite tests. The target is a conjecture in a specified source version; no claim of first discovery or exhaustive verification of present-day open status is made. Classical Christoffel formulas and the known general rationality theory are credited, rather than presented as new.

**Reproducibility.** The accompanying archive contains this L<sup>A</sup>T<sub>E</sub>X source, the PDF, a standard-library-only Python implementation, an independently organized verification suite, and exact machine-readable data. No external mathematical software is needed to rerun the tests.

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## 1 The problem and the contribution

Let

$$C_n = \frac{1}{n+1} \binom{2n}{n} \quad (n \in \mathbb{N}_0) \quad (1.1)$$

be the Catalan numbers. We use determinant size, rather than the last row index, throughout:

$$D_N^{(m)}(a, b) = \det_{0 \leq i, j < N} (aC_{i+j+m} + bC_{i+j+m+1}), \quad D_0^{(m)} = 1. \quad (1.2)$$

Here  $m, N \in \mathbb{N}_0$ , and the parameters may be indeterminates or complex numbers. Define the ordinary generating function

$$F_m(t; a, b) = \sum_{N \geq 0} D_N^{(m)}(a, b) t^N. \quad (1.3)$$

All generating-function identities below first make sense as identities of formal power series. Assertions about poles concern the associated rational functions.

Barry's preprint [1], version 1, explicitly conjectures on page 4 that the exponent of the quadratic generating-function denominator is  $\binom{m}{2} + 1$ . The same page contains a more complicated coefficient conjecture. These are distinct statements: the denominator claim is correct in the uniform sense proved here, whereas the printed coefficient formula uses a shift one smaller than its accompanying determinant. Appendix A makes the translation precise.

There is substantial prior theory. Dougherty, French, Saderholm, and Qian [2] studied recurrences for linear combinations of Catalan numbers. Krattenthaler [3] gives a detailed account of the Christoffel-type determinant identity, confluence, and constant-coefficient recurrences; his Section 8 also records Elouafi's prior proof of general Catalan recurrence assertions. For a multiplier of degree  $r$ , that general construction supplies an order at most  $2^r$ . Consequently, rationality by itself is *not* the research claim here. For the special multiplier  $x^m(a + bx)$ , the objective is the exact, much smaller order, together with its sharp parameter classification.

The article separates two useful calculations. Jacobi orthogonality immediately exposes the individual coefficients. A confluent determinant, on the other hand, exposes the dependence on  $N$  and explains the precise recurrence order. Their independent implementations provide a useful check against indexing and normalization errors.

### 1.1 A notation guide

We shall repeatedly use

$$d = \binom{m}{2}, \quad e = d + 1, \quad K = (m - 1)^2, \quad q(t) = 1 - (a + 2b)t + b^2 t^2. \quad (1.4)$$

The shifted pure Catalan determinant and its leading coefficient are

$$H_N^{(m)} = \det_{0 \leq i, j < N} (C_{i+j+m}) = \prod_{1 \leq j \leq i < m} \frac{2N + i + j}{i + j}, \quad (1.5)$$

$$h_m = \prod_{1 \leq j \leq i < m} \frac{2}{i + j}. \quad (1.6)$$

Every empty product and empty determinant equals 1. In particular,  $H_N^{(0)} = H_N^{(1)} = 1$  and  $h_0 = h_1 = 1$ . We prove (1.5) rather than assuming it. As a polynomial in  $N$ ,  $H_N^{(m)}$  has degree  $d$  and leading coefficient  $h_m$ . The notation  $(u)_k = u(u + 1) \cdots (u + k - 1)$  denotes a rising factorial, with  $(u)_0 = 1$ .

## 2 Statements of the main theorems

**Theorem 2.1** (Explicit coefficients). *For every  $N, m \in \mathbb{N}_0$ ,*

$$D_N^{(m)}(a, b) = \sum_{k=0}^N c_{N,m,k} a^k b^{N-k}, \quad c_{N,m,k} = H_N^{(m+1)} \binom{N}{k} \frac{(N+m+1)_k}{4^k (m+\frac{1}{2})_k}. \quad (2.1)$$

*All the coefficients  $c_{N,m,k}$  are positive integers. Equivalently,*

$$c_{N,m,k} = H_N^{(m+1)} \binom{N}{k} \frac{(N+m+k)!(2m)!(m+k)!}{(N+m)!(2m+2k)!m!}. \quad (2.2)$$

*They can be generated without evaluating any determinant:*

$$c_{N,m,0} = H_N^{(m+1)}, \quad \frac{c_{N,m,k+1}}{c_{N,m,k}} = \frac{(N-k)(N+m+1+k)}{2(k+1)(2m+2k+1)}. \quad (2.3)$$

**Theorem 2.2** (Uniform rational form and sharp denominator). *For every  $m \in \mathbb{N}_0$  there is a polynomial  $P_m \in \mathbb{Z}[a, b, t]$  such that*

$$F_m(t; a, b) = \frac{P_m(t; a, b)}{q(t)^{d+1}}. \quad (2.4)$$

*It satisfies  $P_m(0; a, b) = 1$  and  $\deg_t P_m \leq K$ . If  $b \neq 0$ , then  $\deg_t P_m = K$  and*

$$P_m(t; a, b) = s_m (bt)^K P_m\left(\frac{1}{b^2 t}; a, b\right), \quad s_m = (-1)^{(m-1)(m-2)/2}. \quad (2.5)$$

*In particular,  $[t^K]P_m = s_m b^K$ . If  $b \neq 0$ ,  $a \neq 0$ , and  $a \neq -4b$ , the representation (2.4) is reduced over  $\mathbb{C}[t]$ . The sequence has minimal constant-coefficient recurrence characteristic polynomial*

$$(X^2 - (a+2b)X + b^2)^{d+1}, \quad (2.6)$$

*and hence minimal order  $2d+2 = m(m-1)+2$ .*

Here “minimal” refers even to recurrences allowed to start after a finite initial segment. Thus minimality is not an artifact of the empty-determinant convention. The polynomial numerator in (2.4) remains the *uniform, possibly unreduced* numerator at exceptional parameters.

**Theorem 2.3** (Every exceptional denominator). *For fixed complex  $a, b$ , the reduced denominator of  $F_m$ , normalized to have constant term 1, is exactly the following:*

| Parameter condition        | Reduced denominator           |
|----------------------------|-------------------------------|
| $a = b = 0$                | 1                             |
| $b = 0, a \neq 0$          | $(1 - at)^{\binom{m}{2}+1}$   |
| $a = 0, b \neq 0$          | $(1 - bt)^{\binom{m+1}{2}+1}$ |
| $a = -4b, b \neq 0$        | $(1 + bt)^{\binom{m}{2}+2}$   |
| $b \neq 0, a(a+4b) \neq 0$ | $q(t)^{\binom{m}{2}+1}$       |

**Theorem 2.4** (Finite asymptotic expansion). *Let  $a, b > 0$  be fixed and put*

$$r = \frac{a/b + 2 + \sqrt{(a/b)(a/b+4)}}{2} > 1, \quad \lambda_+ = br, \quad \lambda_- = b/r. \quad (2.7)$$

There are explicit degree- $d$  polynomials  $B_+(N), B_-(N)$  such that

$$D_N^{(m)}(a, b) = B_+(N)\lambda_+^N + B_-(N)\lambda_-^N \quad (N \geq 0). \quad (2.8)$$

In particular,

$$D_N^{(m)}(a, b) = A_m N^d \lambda_+^N \left( 1 + \frac{d \left( \frac{m}{2} - \frac{1}{r-1} \right)}{N} + O(N^{-2}) + O(r^{-2N}) \right), \quad (2.9)$$

$$A_m = h_m \frac{r^{m+1}}{(r-1)^m(r+1)}. \quad (2.10)$$

The entire algebraic expansion terminates after  $N^{-d}$ ; the second exponential is also multiplied by a polynomial of degree  $d$ . For  $d = 0$ , there are no algebraic correction terms at all.

## 2.1 The scale of the improvement

For a degree- $(m+1)$  multiplier the general  $2^{m+1}$  construction need not be minimal. The theorem determines the actual order in this family. The first values are:

| $m$ | Exponent $e$ | Exact order $2e$ | Numerator degree $K$ | $2^{m+1}$ |
|-----|--------------|------------------|----------------------|-----------|
| 0   | 1            | 2                | 1                    | 2         |
| 1   | 1            | 2                | 0                    | 4         |
| 2   | 2            | 4                | 1                    | 8         |
| 3   | 4            | 8                | 4                    | 16        |
| 4   | 7            | 14               | 9                    | 32        |
| 5   | 11           | 22               | 16                   | 64        |
| 6   | 16           | 32               | 25                   | 128       |
| 8   | 29           | 58               | 49                   | 512       |
| 10  | 46           | 92               | 81                   | 2048      |

The reduction is not obtained by fitting a shorter recurrence to a long initial segment. It comes from a cancellation of polynomial degree in a family of minors, proved in Section 5.

## 3 Moments, orthogonality, and the coefficient formula

### 3.1 Catalan moments and a normalized Jacobi polynomial

For  $m \in \mathbb{N}_0$  define a positive measure on  $(0, 4)$  by

$$d\mu_m(x) = \frac{1}{2\pi} x^{m-1/2} (4-x)^{1/2} dx. \quad (3.1)$$

The beta integral, after  $x = 4t$ , gives

$$\int_0^4 x^n d\mu_m(x) = \frac{4^{n+m+1}}{2\pi} B(n+m+\frac{1}{2}, \frac{3}{2}) = C_{n+m}. \quad (3.2)$$

Consequently every moment matrix is positive definite, so its determinant is nonzero and its monic orthogonal polynomials are well-defined.

**Lemma 3.1** (The monic polynomial). *The degree- $N$  monic orthogonal polynomial for  $\mu_m$  is*

$$p_N^{(m)}(x) = (-1)^N \frac{4^N (m + \frac{1}{2})_N}{(N + m + 1)_N} {}_2F_1 \left( \begin{matrix} -N, N + m + 1 \\ m + \frac{1}{2} \end{matrix}; x/4 \right). \quad (3.3)$$

*The hypergeometric series terminates after  $N + 1$  terms.*

*Proof.* Write  $\alpha = m - \frac{1}{2}$ ,  $\beta = \frac{1}{2}$ , so both are greater than  $-1$ . A Rodrigues representation on  $0 < t < 1$  is

$$\frac{t^{-\alpha}(1-t)^{-\beta}}{(\alpha+1)_N} \frac{d^N}{dt^N} \left( t^{N+\alpha}(1-t)^{N+\beta} \right) = {}_2F_1 \left( \begin{matrix} -N, N + \alpha + \beta + 1 \\ \alpha + 1 \end{matrix}; t \right). \quad (3.4)$$

Leibniz's rule gives the polynomial on the right, including its normalization at  $t = 0$ . Multiplying the left side by  $t^j(1-t)^\beta$  and integrating by parts  $N$  times yields zero for  $0 \leq j < N$ . Every boundary term vanishes: after at most  $N - 1$  derivatives, the endpoint exponents that can occur are at least  $\alpha + 1 > 0$  and  $\beta + 1 > 0$ . The final derivative of  $t^j$  is zero.

The coefficient of  $t^N$  on the right is  $(-1)^N (N + \alpha + \beta + 1)_N / (\alpha + 1)_N$ . Replacing  $t$  by  $x/4$  and normalizing this leading coefficient gives (3.3). This also identifies the polynomial with the standard Jacobi family; see the conventional hypergeometric representation in DLMF [4], Eq. 18.5.7.  $\square$

### 3.2 A characteristic-polynomial determinant identity

The following elementary identity will be used twice.

**Lemma 3.2.** *Let  $v_j$  be the moments of a positive measure, let  $H_N = \det(v_{i+j})_{0 \leq i, j < N}$ , and let  $p_N$  be its monic orthogonal polynomial. Then*

$$\det_{0 \leq i, j < N} (v_{i+j+1} - z v_{i+j}) = (-1)^N H_N p_N(z). \quad (3.5)$$

*Proof.* The monic polynomial is represented by

$$p_N(z) = \frac{1}{H_N} \det \begin{pmatrix} v_0 & v_1 & \cdots & v_N \\ v_1 & v_2 & \cdots & v_{N+1} \\ \vdots & \vdots & & \vdots \\ v_{N-1} & v_N & \cdots & v_{2N-1} \\ 1 & z & \cdots & z^N \end{pmatrix}.$$

Its leading coefficient is one, and integrating it against any power  $z^j$ ,  $j < N$ , creates a repeated row. Thus it is indeed  $p_N$ . Starting with the last column and moving left, replace  $\text{col}_j$  by  $\text{col}_j - z \text{col}_{j-1}$ . The last row becomes  $(1, 0, \dots, 0)$ . Expansion along it gives (3.5), with sign  $(-1)^N$ .  $\square$

Applying the lemma to  $\mu_m$  and  $z = -a/b$  gives, for  $b \neq 0$ ,

$$D_N^{(m)}(a, b) = (-b)^N H_N^{(m)} p_N^{(m)}(-a/b). \quad (3.6)$$

### 3.3 A proof of the shifted Catalan product

For  $m = 0$ , (3.3) reduces to

$$p_n^{(0)}(x) = \sum_{j=0}^n (-1)^{n-j} \binom{n+j}{2j} x^j. \quad (3.7)$$

Its initial values and recurrence are

$$p_0^{(0)} = 1, \quad p_1^{(0)} = x - 1, \quad p_{n+2}^{(0)} = (x - 2)p_{n+1}^{(0)} - p_n^{(0)}. \quad (3.8)$$

These follow directly by comparing the binomial coefficients in (3.7). If  $\eta_n$  denotes the squared norm of  $p_n^{(0)}$ , the coefficient 1 of  $p_{n-1}^{(0)}$  in  $x p_n^{(0)}$  implies  $\eta_n = \eta_{n-1}$ . Indeed, the two ways of writing  $\langle x p_n^{(0)}, p_{n-1}^{(0)} \rangle$  give  $\eta_{n-1}$  and  $\eta_n$ , respectively. Since  $\eta_0 = C_0 = 1$ , all these norms equal 1. Changing from monomials to monic orthogonal polynomials is unitriangular, and therefore  $H_N^{(0)} = \prod_{n=0}^{N-1} \eta_n = 1$ .

Next set  $z = 0$  in (3.5) and use (3.3). We obtain

$$\frac{H_N^{(m+1)}}{H_N^{(m)}} = (-1)^N p_N^{(m)}(0) = \frac{4^N (m + \frac{1}{2})_N}{(N + m + 1)_N} = \prod_{j=1}^m \frac{2N + m + j}{m + j}. \quad (3.9)$$

For completeness, the middle expression equals

$$\frac{(2m + 2N)! m! (N + m)!}{(m + N)! (2m)! (2N + m)!} = \frac{(2N + 2m)! m!}{(2m)! (2N + m)!},$$

which is the last product. Induction on  $m$ , beginning with  $H_N^{(0)} = 1$ , proves (1.5).

### 3.4 Completion of the coefficient proof

Insert (3.3) into (3.6). The constant factor is  $H_N^{(m+1)}$  by (3.9). In each summand,  $(-N)_k (-1)^k / k! = \binom{N}{k}$ . This gives exactly (2.1). The formula is a polynomial identity for  $b \neq 0$ , so it holds for  $b = 0$  as well. The duplication identity for a rising half-integer factorial, or direct cancellation, gives (2.2). Taking adjacent coefficient ratios gives (2.3).

All factors in (2.1) are positive rational numbers. The left side is a determinant with entries in  $\mathbb{Z}[a, b]$ , so its coefficients are integers. This proves positivity and integrality simultaneously and completes Theorem 2.1. In particular,

$$c_{N,m,0} = H_N^{(m+1)}, \quad c_{N,m,N} = H_N^{(m)}. \quad (3.10)$$

### 3.5 Real roots and coefficient inequalities

**Corollary 3.3.** *For  $N \geq 1$ , the polynomial  $a \mapsto D_N^{(m)}(a, 1)$  has exactly  $N$  simple real roots, all in  $(-4, 0)$ . For  $N \geq 2$ , its normalized coefficients  $u_k = c_{N,m,k} / \binom{N}{k}$  satisfy*

$$u_k^2 > u_{k-1} u_{k+1} \quad (1 \leq k < N). \quad (3.11)$$

*Proof.* A degree- $N$  orthogonal polynomial for a strictly positive density on  $(0, 4)$  has  $N$  simple roots in that interval. One elementary proof is to multiply it by the product of its distinct sign-changing roots in  $(0, 4)$ . If there were fewer than  $N$  such roots, that product would have degree below  $N$ , while the integral of its product with the orthogonal polynomial would have constant nonzero sign. This contradicts orthogonality. Equation (3.6) now negates the roots.

Alternatively, the coefficient inequality follows without any root theory. Theorem 2.1 gives

$$\frac{u_{k+1}}{u_k} = \frac{N + m + 1 + k}{4(m + \frac{1}{2} + k)}.$$

This is strictly decreasing in  $k$ , because the numerator's constant term exceeds the denominator's unscaled constant term by  $N + 1/2$ . The decrease is equivalent to (3.11).  $\square$

#### 4 A fixed-size confluent determinant

The explicit coefficient formula grows in length with  $N$ . To understand the recurrence order, we need a representation whose determinant size depends only on  $m$ . Define

$$R_n(y) = (-1)^n p_n^{(0)}(-y) = \sum_{i=0}^n \binom{n+i}{2i} y^i. \quad (4.1)$$

Thus

$$R_0(y) = 1, \quad R_1(y) = y + 1, \quad R_{n+2}(y) = (y + 2)R_{n+1}(y) - R_n(y). \quad (4.2)$$

Extend  $R_n$  to all integer indices using this recurrence. Since  $R_{-1} = R_0 = 1$ , it also satisfies the reflection identity

$$R_{-n-1}(y) = R_n(y) \quad (n \in \mathbb{Z}). \quad (4.3)$$

The reversed form of the polynomials is convenient because it eliminates unnecessary signs from Christoffel's identity.

**Lemma 4.1** (Christoffel determinant in the present normalization). *For pairwise distinct  $y_1, \dots, y_r$ ,*

$$\det_{0 \leq i, j < N} \int_0^4 x^{i+j} \prod_{\ell=1}^r (x + y_\ell) d\mu_0(x) = \frac{\det[R_{N+j}(y_i)]_{\substack{1 \leq i \leq r \\ 0 \leq j < r}}}{\prod_{1 \leq i < j \leq r} (y_j - y_i)}. \quad (4.4)$$

*The quotient extends polynomially when some  $y_i$  coincide.*

*Proof.* Here is a proof that also verifies the normalization. Write  $\Delta(x_1, \dots, x_N) = \det[x_i^{j-1}]_{1 \leq i, j \leq N}$ . Expanding two determinants and integrating termwise gives the identity

$$\frac{1}{N!} \int \cdots \int \det[f_i(x_j)] \det[g_i(x_j)] \prod_j d\mu_0(x_j) = \det \left[ \int f_i(x) g_j(x) d\mu_0(x) \right].$$

No analytic interchange difficulty occurs here: the sums are finite and all polynomials are integrable on a bounded interval.

For external variables  $z_1, \dots, z_r$  we have

$$\Delta(x, z) = \Delta(x) \Delta(z) \prod_{i=1}^N \prod_{\ell=1}^r (z_\ell - x_i).$$

In the determinant defining  $\Delta(x, z)$ , replace monomials by the monic polynomials  $p_j^{(0)}$ ; the determinant is unchanged. Expand along its first  $N$  rows and integrate against  $\Delta(x)$ . By orthogonality, every choice of columns vanishes except columns of degrees  $0, \dots, N-1$ . That surviving integral equals  $N! H_N^{(0)} = N!$ . The remaining minor is  $\det[p_{N+j}^{(0)}(z_i)]$ . Dividing by  $\Delta(z)$  yields the identity for the multiplier  $\prod_\ell (z_\ell - x)$ .

Finally put  $z_\ell = -y_\ell$  and use  $p_{N+j}^{(0)}(-y) = (-1)^{N+j} R_{N+j}(y)$ . The signs from the multiplier, column factors, and Vandermonde cancel, giving (4.4). The numerator is alternating in the  $y_i$ , hence divisible by their Vandermonde as a polynomial. This proves the extension statement. This is the classical fixed-size identity discussed in [3]; its inclusion here is a normalization proof, not a novelty claim.  $\square$

Take  $r = m + 1$ , let the first  $m$  variables tend to zero, and put the last one equal to  $y = a/b \neq 0$ . Dividing by the Vandermonde and taking the limit replaces those first rows by Taylor coefficients. Since  $[y^i] R_{N+j}(y) = \binom{N+j+i}{2i}$ , we obtain:

**Proposition 4.2** (Confluent representation). *If  $ab \neq 0$ , then*

$$D_N^{(m)}(a, b) = b^N y^{-m} \det \begin{pmatrix} \binom{N+j+i}{2i} & (0 \leq i < m, 0 \leq j \leq m) \\ R_{N+j}(y) & (0 \leq j \leq m) \end{pmatrix}, \quad y = a/b. \quad (4.5)$$

The upper block has  $m$  rows and the lower block has one row. For  $m = 0$  the formula simply reads  $D_N^{(0)} = b^N R_N(a/b)$ .

Define the  $m \times m$  minors

$$M_{m,\ell}(N) = \det_{\substack{0 \leq i < m \\ j \in J_\ell}} \binom{N+j+i}{2i}, \quad J_\ell = \{0, 1, \dots, m\} \setminus \{\ell\}, \quad (4.6)$$

where columns are in increasing order and  $0 \leq \ell \leq m$ . Expansion of (4.5) along its last row gives

$$D_N^{(m)}(a, b) = b^N y^{-m} \sum_{\ell=0}^m (-1)^{m+\ell} M_{m,\ell}(N) R_{N+\ell}(y). \quad (4.7)$$

Although a naive degree bound for  $M_{m,\ell}$  is  $m(m-1) = 2d$ , its actual degree is only  $d$ . This is the key point.

## 5 The degree cancellation and its first correction

**Lemma 5.1** (Sharp minor degree). *For  $0 \leq \ell \leq m$ ,  $M_{m,\ell}(N)$  is a polynomial of degree  $d = \binom{m}{2}$  with leading coefficient*

$$[N^d]M_{m,\ell}(N) = h_m \binom{m}{\ell}. \quad (5.1)$$

For  $d \geq 1$ , the next coefficient is

$$[N^{d-1}]M_{m,\ell}(N) = h_m \binom{m}{\ell} \left( \frac{d(m+2)}{2} - \frac{(m-1)\ell}{2} \right). \quad (5.2)$$

*Proof.* Let  $f_i(x) = \binom{x+i}{2i}$ , so  $f_i$  has degree  $2i$  and leading coefficient  $1/(2i)!$ . The determinant  $\det[f_i(x_j)]_{0 \leq i, j < m}$  is alternating in its  $m$  independent variables. It is divisible by  $\Delta(x) = \prod_{p < q} (x_q - x_p)$ , whose degree is  $d$ . The determinant has total degree at most  $\sum_i 2i = 2d$ ; its quotient by  $\Delta$  therefore has total degree at most  $d$ . Substituting  $x_j = N + j$  for  $j \in J_\ell$  makes  $\Delta$  a nonzero constant. Hence the degree in  $N$  is at most  $d$ .

To obtain the leading coefficient, expand  $f_i(N + j)$  in Taylor powers of  $j$ . The first possible nonzero determinant uses distinct Taylor orders  $0, 1, \dots, m-1$ ; these have the least possible sum  $d$ . The coefficient of  $N^d$  is

$$\frac{\det[\binom{2i}{r}]_{0 \leq i, r < m}}{\prod_{i=0}^{m-1} (2i)!} \Delta(J_\ell) = \frac{2^d \Delta(J_\ell)}{\prod_{i=0}^{m-1} (2i)!}. \quad (5.3)$$

To see the second equality, regard  $\binom{2i}{r}$  as a polynomial of degree  $r$  in  $i$  with leading coefficient  $2^r/r!$  and evaluate its determinant at  $i = 0, \dots, m-1$ . The Vandermonde is  $\prod_{r=1}^{m-1} r!$ , so these factorials cancel. Removing  $\ell$  from  $0, \dots, m$  also gives

$$\Delta(J_\ell) = \binom{m}{\ell} \prod_{r=1}^{m-1} r!. \quad (5.4)$$

Since

$$h_m = \frac{2^d \prod_{r=1}^{m-1} r!}{\prod_{i=0}^{m-1} (2i)!} \quad (m \geq 1),$$

we obtain (5.1). It is nonzero, proving the degree assertion. The case  $m = 0$  is the empty determinant and is immediate.

For the next coefficient, observe the symmetric factorization

$$f_i(x) = \frac{1}{(2i)!} \prod_{r=1}^i \left( \left(x + \frac{1}{2}\right)^2 - \left(r - \frac{1}{2}\right)^2 \right). \quad (5.5)$$

Thus  $f_i(x) = (x + \frac{1}{2})^{2i} / (2i)!$  plus terms of degree at most  $2i - 2$ . Replacing one or more rows by those lower-degree terms reduces the total degree by at least two. The same alternating argument then bounds their degree after translation by  $d - 2$ . The leading two terms are consequently obtained from

$$\det \left[ \frac{(N + j + \frac{1}{2})^{2i}}{(2i)!} \right]_{0 \leq i < m, j \in J_\ell} = \frac{\prod_{\substack{p < q \\ p, q \in J_\ell}} (q - p)(2N + p + q + 1)}{\prod_{i=0}^{m-1} (2i)!}. \quad (5.6)$$

The relative coefficient of  $N^{d-1}$  is

$$\frac{1}{2} \sum_{\substack{p < q \\ p, q \in J_\ell}} (p + q + 1) = \frac{1}{2} \left( (m-1) \left( \frac{m(m+1)}{2} - \ell \right) + d \right),$$

which simplifies to (5.2). If  $d = 1$ , the putative remainder of degree  $d - 2 = -1$  is identically zero.  $\square$

It is useful to collect the minors in a polynomial in an auxiliary variable  $r$ :

$$Q_m(N; r) = \sum_{\ell=0}^m (-1)^{m+\ell} M_{m,\ell}(N) r^\ell. \quad (5.7)$$

By the binomial theorem,

$$[N^d] Q_m(N; r) = h_m(r - 1)^m. \quad (5.8)$$

When  $r \neq 1$  and  $d \geq 1$ , the relative next coefficient is

$$\frac{[N^{d-1}] Q_m(N; r)}{[N^d] Q_m(N; r)} = \frac{d(m+2)}{2} - d \frac{r}{r-1} = d \left( \frac{m}{2} - \frac{1}{r-1} \right). \quad (5.9)$$

Indeed, applying  $r \, d/dr$  to  $(r - 1)^m$  sums the terms weighted by  $\ell$  in (5.2). The exact value  $d$ , rather than an upper bound  $2d$ , is what makes the denominator exponent sharp.

## 6 Two exponentials and the minimal recurrence

Suppose first that  $b \neq 0$  and  $a(a + 4b) \neq 0$ . Set  $y = a/b$  and choose one ordering of the two roots

$$r_{\pm} = \frac{y + 2 \pm \sqrt{y(y+4)}}{2}, \quad r_+ r_- = 1. \quad (6.1)$$

They are distinct and nonzero. Solving the recurrence (4.2) with its two initial values gives

$$R_n(y) = \frac{(r_+ - 1)r_+^n - (r_- - 1)r_-^n}{r_+ - r_-}. \quad (6.2)$$

Write

$$\lambda_{\pm} = b r_{\pm} = \frac{a + 2b \pm \sqrt{a(a+4b)}}{2}. \quad (6.3)$$

Substitution in (4.7) gives the exact identity

$$D_N^{(m)}(a, b) = \frac{(r_+ - 1)Q_m(N; r_+)\lambda_+^N - (r_- - 1)Q_m(N; r_-)\lambda_-^N}{y^m(r_+ - r_-)}. \quad (6.4)$$

In particular, both polynomial coefficients in this two-exponential representation have degree *exactly*  $d$ : neither  $r_+$  nor  $r_-$  can equal 1, since  $y \neq 0$ .

**Lemma 6.1** (Minimal annihilator of an exponential polynomial). *If  $\lambda_1, \dots, \lambda_s$  are distinct nonzero complex numbers and the nonzero polynomials  $A_i(N)$  have degrees  $d_i$ , then  $u_N = \sum_i A_i(N)\lambda_i^N$  has minimal constant-coefficient annihilator*

$$\prod_{i=1}^s (E - \lambda_i)^{d_i+1}, \quad Eu_N = u_{N+1}. \quad (6.5)$$

*Its generating function has poles of orders  $d_i + 1$  at  $t = 1/\lambda_i$ .*

*Proof.* On  $A(N)\lambda^N$ , the operator  $E - \lambda$  is  $\lambda^{N+1}(A(N+1) - A(N))$ ; it lowers the polynomial degree by exactly one until the result vanishes. For  $\mu \neq \lambda$ ,  $E - \mu$  preserves that degree, multiplying the leading coefficient by  $\lambda - \mu$ . The generalized eigenspaces for distinct eigenvalues are independent: polynomial projectors from the Chinese remainder identity isolate each one. Thus every annihilator needs the stated multiplicity at every eigenvalue. This argument remains valid after any finite shift in  $N$ .

Equivalently, applying  $(z d/dz)^k$  to  $1/(1-z)$  shows that  $\sum_{N \geq 0} N^k z^N$  has a pole of exact order  $k+1$  at  $z=1$ , with nonzero leading pole coefficient  $k!$ . Distinct pole locations cannot cancel. This proves the generating-function assertion too.  $\square$

Applying the lemma to (6.4) proves the minimal characteristic polynomial (2.6). It also proves that  $q^{d+1}F_m$  is a polynomial of degree at most  $2d+1 = 2e-1$  for these parameters. Section 7 sharpens its degree to  $K$ .

## 6.1 Why the identity extends to all parameters

Each  $D_N^{(m)}$  is a polynomial in  $a, b$  with integer coefficients. The coefficients of  $q^{d+1}F_m$  are therefore such polynomials too. Every coefficient that vanishes for all  $b \neq 0$  and  $a(a+4b) \neq 0$  vanishes identically: a polynomial cannot be nonzero and vanish on that nonempty open set. Hence the uniform rational identity, and its coefficient integrality, extend to every  $a, b$ , even where its displayed denominator is not reduced. This argument is coefficientwise in a formal series and does not require a convergence limit in  $t$ .

## 6.2 Proof of the exceptional classification

When  $b = 0$ ,

$$D_N^{(m)}(a, 0) = a^N H_N^{(m)}. \quad (6.6)$$

For  $a \neq 0$ , the polynomial  $H_N^{(m)}$  has degree  $d$ , so Lemma 6.1 gives denominator  $(1-at)^{d+1}$ . When  $a = 0$  and  $b \neq 0$ ,

$$D_N^{(m)}(0, b) = b^N H_N^{(m+1)}, \quad (6.7)$$

whose polynomial degree is  $\binom{m+1}{2}$ .

At the remaining collision  $a = -4b$ , with  $b \neq 0$ , the recurrence gives

$$R_n(-4) = (-1)^n (2n+1). \quad (6.8)$$

Inserting this into (4.7) and tracking the signs produces

$$D_N^{(m)}(-4b, b) = (-b)^N E_m(N), \quad (6.9)$$

$$E_m(N) = 4^{-m} \sum_{\ell=0}^m (2N + 2\ell + 1) M_{m,\ell}(N). \quad (6.10)$$

By (5.1), its degree is exactly  $d + 1$ , with

$$[N^{d+1}]E_m(N) = 4^{-m} 2h_m \sum_{\ell=0}^m \binom{m}{\ell} = 2^{1-m} h_m \neq 0. \quad (6.11)$$

Thus the reduced denominator is  $(1 + bt)^{d+2}$ . Finally, if  $a = b = 0$ , every nonempty determinant vanishes and  $F_m = 1$ . Together with the nonexceptional result, this proves Theorem 2.3.

The two root-collision lines have different pole orders. This is not a contradiction:  $x = 0$  and  $x = 4$  are inequivalent endpoints of the weight  $x^{m-1/2}(4-x)^{1/2}$ . In particular, blindly substituting  $a = 0$  into a generic minimal-order statement would be incorrect.

## 7 The numerator degree and reciprocal symmetry

A rational function can have a denominator of large degree and a surprisingly short numerator. Here the explanation is a string of zero values in the natural bilateral extension of the determinant.

For the moment retain the nonexceptional hypotheses. Define  $D_N$  for all  $N \in \mathbb{Z}$  by the right side of (4.5). The upper rows are interpreted through their polynomial binomial coefficients; this agrees with the Taylor coefficients of  $R_n$  at zero, including negative  $n$ , because

$$\binom{-n-1+i}{2i} = \binom{n+i}{2i}.$$

It also agrees with the unique bilateral extension supplied by (6.4).

**Lemma 7.1** (Reflection and a zero interval). *Let  $\sigma_m = (-1)^{\binom{m+1}{2}}$ . Then*

$$D_{-N-m-1} = \sigma_m b^{-2N-m-1} D_N \quad (N \in \mathbb{Z}), \quad (7.1)$$

$$D_{-1} = D_{-2} = \cdots = D_{-m} = 0. \quad (7.2)$$

*The second statement is empty for  $m = 0$ .*

*Proof.* The index change  $N \mapsto -N - m - 1$ , followed by  $R_{-n-1} = R_n$ , reverses all  $m + 1$  columns of the determinant. The sign of that reversal is  $\sigma_m$ . Comparing its prefactor  $b^N$  on the two sides gives (7.1). If  $N = -s$  with  $1 \leq s \leq m$ , columns  $j = s - 1$  and  $j = s$  correspond to  $R_{-1}$  and  $R_0$ , respectively, which are the same polynomial. Their values and all needed Taylor coefficients coincide, so the determinant vanishes.  $\square$

**Lemma 7.2** (The expansion at infinity). *For a sequence of exponential-polynomial form with nonzero characteristic roots and its bilateral extension, its proper rational generating function satisfies*

$$F(t) = - \sum_{r \geq 1} D_{-r} t^{-r} \quad (7.3)$$

*as a formal Laurent expansion at infinity.*

*Proof.* For  $D_N = \lambda^N$  the assertion is simply  $(1 - \lambda t)^{-1} = - \sum_{r \geq 1} \lambda^{-r} t^{-r}$ . Apply powers of  $t \, d/dt$  to obtain the assertion for  $D_N = N^k \lambda^N$ , and then use linearity.  $\square$

Combining (7.2), (7.1), and  $D_0 = 1$  yields

$$F_m(t) = -\sigma_m b^{-m-1} t^{-m-1} + O(t^{-m-2}) \quad (t \rightarrow \infty). \quad (7.4)$$

Since  $q^e$  has degree  $2e$  and leading coefficient  $b^{2e}$ , the numerator has degree

$$2e - m - 1 = m(m - 1) + 2 - m - 1 = (m - 1)^2 = K \quad (7.5)$$

and leading coefficient  $-\sigma_m b^{2e-m-1} = s_m b^K$ . The identity  $s_m = -\sigma_m$  follows because the difference of the two exponents  $\binom{m+1}{2}$  and  $(m - 1)(m - 2)/2$  is  $2m - 1$ , an odd integer. This calculation proves the sharp degree assertion, including  $m = 0$ .

The full reflection identity yields more than the leading term. Putting  $1/(b^2 t)$  into (7.3) and reindexing with (7.1), we find

$$F_m\left(\frac{1}{b^2 t}\right) = s_m b^{m+1} t^{m+1} F_m(t). \quad (7.6)$$

But  $q(1/(b^2 t)) = q(t)/(b^2 t^2)$ . Substitution into (7.6) gives exactly (2.5). After denominators in  $b$  are cleared, each of these numerator identities is polynomial in  $a, b$ . Thus they extend to the exceptional parameters; for  $b \neq 0$  the leading coefficient remains nonzero. This completes the proof of Theorem 2.2.

### 7.1 An explicit finite prescription for the numerator

No uncomputed infinite series is needed to determine  $P_m$ . Expand

$$q(t)^e = \sum_{j=0}^{2e} q_j t^j. \quad (7.7)$$

For example, its coefficients have the finite expression

$$q_j = \sum_{v=\max(0, j-e)}^{\lfloor j/2 \rfloor} \frac{e!}{(j-2v)! v! (e-j+v)!} (-(a+2b))^{j-2v} b^{2v}. \quad (7.8)$$

Then

$$P_m(t) = \sum_{\ell=0}^K \left( \sum_{j=0}^{\min(\ell, 2e)} q_j D_{\ell-j}^{(m)}(a, b) \right) t^\ell, \quad (7.9)$$

where every determinant in the right side is given by Theorem 2.1. Equations (7.8) and (7.9) therefore constitute a finite sum/product formula for every numerator coefficient. All coefficients of  $q^e F_m$  after degree  $K$  vanish.

The usual recurrence is

$$\sum_{j=0}^{2e} q_j D_{N-j}^{(m)}(a, b) = 0 \quad (N \geq 2e). \quad (7.10)$$

In fact the truncated convolution already vanishes for  $N > K$  when negative-index terms are taken as zero for the purpose of multiplying ordinary power series. Those artificial zeros should not be confused with the distinct bilateral extension used in the proof above.

## 8 Exact exponential expansions and asymptotics

Equation (6.4) supplies every coefficient of the two polynomials in Theorem 2.4. Explicitly,

$$B_+(N) = \frac{(r_+ - 1)Q_m(N; r_+)}{y^m(r_+ - r_-)}, \quad B_-(N) = -\frac{(r_- - 1)Q_m(N; r_-)}{y^m(r_+ - r_-)}. \quad (8.1)$$

These formulas are valid for arbitrary nonexceptional complex parameters; the assumption  $a, b > 0$  is only needed to select a strictly dominant real exponential for the asymptotic statement.

For  $a, b > 0$ , write  $r = r_+$ , so  $r_- = r^{-1}$  and

$$y = r + r^{-1} - 2 = \frac{(r-1)^2}{r}. \quad (8.2)$$

Using (5.8), the coefficient of  $N^d$  in  $B_+$  is

$$\frac{h_m(r-1)^{m+1}}{y^m(r-r^{-1})} = h_m \frac{r^{m+1}}{(r-1)^m(r+1)} = A_m.$$

Its relative next coefficient is exactly (5.9). Also  $\lambda_-/\lambda_+ = r^{-2}$ . Division of (6.4) by  $A_m N^d \lambda_+^N$  proves (2.9).

One may write the complete answer in the finite form

$$D_N^{(m)} = \lambda_+^N \sum_{j=0}^d \beta_j^+ N^{d-j} + \lambda_-^N \sum_{j=0}^d \beta_j^- N^{d-j}, \quad (8.3)$$

where

$$\beta_j^+ = \frac{r_+ - 1}{y^m(r_+ - r_-)} [N^{d-j}] Q_m(N; r_+), \quad \beta_j^- = -\frac{r_- - 1}{y^m(r_+ - r_-)} [N^{d-j}] Q_m(N; r_-). \quad (8.4)$$

Thus the “full asymptotic expansion” is particularly strong here: it is an exact expression, with no uncomputed remainder once both finite sectors are retained. No assertion of asymptotic convergence or summability is needed. The leading coefficient ratio of the two sectors is

$$\frac{\beta_0^-}{\beta_0^+} = (-1)^m r^{-(m+1)}. \quad (8.5)$$

These estimates are for fixed  $m, a, b$ . They are not claimed to be uniform when  $a \rightarrow 0$ , when the two roots coalesce, or when  $m$  grows with  $N$ . In such regimes the exact finite formula remains valid away from the collision itself, but a different uniform asymptotic analysis would be required. At a collision, use the exact exceptional formulas from Section 6 instead.

## 9 Worked examples and integer sequences

### 9.1 Small shifts

The first uniform generating functions are

$$F_0(t) = \frac{1-bt}{q(t)}, \quad F_1(t) = \frac{1}{q(t)}, \quad (9.1)$$

$$F_2(t) = \frac{1+bt}{q(t)^2}, \quad F_3(t) = \frac{(1-b^2t^2)(1+(a+6b)t+b^2t^2)}{q(t)^4}. \quad (9.2)$$

They can be obtained directly from (7.9). Notice that  $m = 0$  is a genuine boundary case: its numerator has degree 1, not 0. For  $m = 3$ , the numerator is anti-reciprocal; for  $m = 2$ , it is reciprocal. This is the pattern described uniformly by  $s_m$ .

For  $m = 2$ , put  $A = a + 2b$ . The recurrence becomes

$$D_N = 2AD_{N-1} - (A^2 + 2b^2)D_{N-2} + 2Ab^2D_{N-3} - b^4D_{N-4} \quad (N \geq 4). \quad (9.3)$$

The initial polynomials are

$$\begin{aligned} D_0 &= 1, \\ D_1 &= 2a + 5b, \\ D_2 &= 3a^2 + 14ab + 14b^2, \\ D_3 &= 4a^3 + 27a^2b + 54ab^2 + 30b^3. \end{aligned}$$

The recurrence is minimal outside the exceptional lines; displaying only a valid order-four relation would not by itself prove that fact.

## 9.2 The specialization $a = b = 1$

The first sequences, all beginning with the empty determinant, are

| $m \backslash N$ | 0 | 1  | 2   | 3    | 4    | 5     | 6      |
|------------------|---|----|-----|------|------|-------|--------|
| 0                | 1 | 2  | 5   | 13   | 34   | 89    | 233    |
| 1                | 1 | 3  | 8   | 21   | 55   | 144   | 377    |
| 2                | 1 | 7  | 31  | 115  | 390  | 1254  | 3893   |
| 3                | 1 | 19 | 170 | 1075 | 5580 | 25529 | 107036 |

The first two are  $F_{2N+1}$  and  $F_{2N+2}$ , respectively, as follows from their common recurrence  $u_{N+2} = 3u_{N+1} - u_N$  and initial values. They connect the family to the odd and even Fibonacci subsequences in OEIS A001519 and A001906 [6]; Catalan numbers themselves are A000108 [5]. The higher rows and their coefficient triangles are natural OEIS-style objects, but the archive does not assert that any particular row is absent from OEIS.

For every fixed shift, the dominant exponential is

$$\lambda_+ = \frac{3 + \sqrt{5}}{2},$$

while its polynomial prefactor has degree  $\binom{m}{2}$ . Thus increasing the shift changes the power of  $N$ , not the exponential growth constant. For  $m = 2$ , the first relative asymptotic correction equals  $1 - 1/(r - 1) = (3 - \sqrt{5})/2$  with  $r = (3 + \sqrt{5})/2$ .

## 9.3 Rational characteristic roots: $a = 1, b = 2$

This specialization gives  $r_+ = 2, r_- = 1/2$  and  $\lambda_+ = 4, \lambda_- = 1$ . Therefore the exact formulas have rational polynomial coefficients and no radicals:

$$D_N^{(0)}(1, 2) = \frac{2 \cdot 4^N + 1}{3}, \quad (9.4)$$

$$D_N^{(1)}(1, 2) = \frac{4 \cdot 4^N - 1}{3}, \quad (9.5)$$

$$D_N^{(2)}(1, 2) = \frac{8N \cdot 4^N + N + 3}{3}, \quad (9.6)$$

$$D_N^{(3)}(1, 2) = \frac{(32N^3 + 48N^2 + 16N + 96)4^N - (2N^3 + 21N^2 + 73N + 78)}{18}. \quad (9.7)$$

For example, the  $m = 2$  sequence starts 1, 12, 87, 514, 2733, 13656, ... Its dominant polynomial is  $(8/3)N$  with no constant term. This vanishing agrees with the first-correction formula  $d(m/2 - 1/(r - 1)) = 1(1 - 1) = 0$ ; the entire correction is the smaller exponential sector  $(N + 3)/3$ . For  $m = 3$ , the relative coefficient of  $N^{-1}$  in the dominant sector is  $3/2$ , again exactly as predicted.

### 9.4 An endpoint example

For  $m = 2$ , (6.10) simplifies to

$$E_2(N) = \frac{(N+1)(N+2)}{2}.$$

Hence

$$D_N^{(2)}(-4b, b) = (-b)^N \frac{(N+1)(N+2)}{2}, \quad F_2(t; -4b, b) = \frac{1}{(1+bt)^3}. \quad (9.8)$$

This also follows by cancelling  $1+bt$  from  $(1+bt)/(1+bt)^4$ , the uniform form of  $F_2$ . At the other collision,  $a = 0$ , one instead obtains  $F_2(t; 0, b) = (1+bt)/(1-bt)^4$ . These explicit cases illustrate why the reduced denominator needs a separate classification.

## 10 Algorithms, exact experiments, and reproducibility

### 10.1 Implementation choices

The file `code/catalan_hankel.py` implements the coefficient formula, pure shifted product, homogeneous polynomial evaluation, fixed-size determinant, and uniform and reduced denominators. All numerical parameters in its public computation interface are integers; internal fractions use Python's exact `Fraction`. The mathematical theorems have the broader complex-parameter scope stated above. No floating-point arithmetic enters the tests.

For a coefficient row, first compute the triangular product  $H_N^{(m+1)}$  and then apply (2.3) successively. The number of arithmetic operations is  $O(m^2 + N)$ ; this is an *arithmetic-operation count*, not a bit-complexity claim. Every division in the update is checked for exactness. Homogeneous Horner evaluation avoids division by  $a$  or  $b$ , so zero parameters are handled without special numerical limits.

The independent determinant path uses a fraction-free Bareiss algorithm on the original matrix of Catalan moments, including pivoting when a diagonal pivot vanishes. This matters for signed parameters: an implementation tested only on positive definite examples can conceal pivot failures. The source-indexing formula is implemented separately, not obtained by calling the corrected coefficient routine.

For many consecutive values at fixed  $m$ , it is also possible to use (7.10) after producing the first  $2e$  values. For very large isolated indices, powering the companion matrix or reducing  $X^N$  modulo (2.6) provides standard recurrence-based methods. These methods are consequences of the theorem; the supplied reference implementation prioritizes transparent exact verification rather than optimized large-integer throughput.

### 10.2 Verification that was actually run

The included `data/verification.json` records PASS for all checks in the following table. The report is generated by the test program, not manually written.

| Check                                                     | Count |
|-----------------------------------------------------------|-------|
| Direct determinant versus coefficient evaluation          | 1,183 |
| Additional direct signed/degenerate parameter evaluations | 810   |
| Printed source formula versus corrected shift             | 990   |
| Confluent fixed-size determinant evaluations              | 364   |
| Minor polynomials exactly interpolated and checked        | 36    |
| Generating-function/recurrence coefficient identities     | 2,816 |
| Numerator reciprocal coefficient identities               | 604   |
| Nonexceptional numerator–denominator gcd checks           | 44    |
| Exceptional reduced-denominator checks                    | 44    |
| Exact two-exponential evaluations                         | 168   |
| Strict normalized log-concavity inequalities              | 5,655 |

The first 1,183 evaluations cover every pair  $0 \leq N, m \leq 12$ , using  $a = 0, 1, \dots, N$  and  $b = 1$ . At each fixed  $(N, m)$ , both sides are polynomials of degree at most  $N$  in  $a$ . Equality at  $N + 1$  distinct points certifies that particular polynomial identity exactly. Thus this part certifies 169 full coefficient polynomials, not merely 169 evaluations. It still does not prove the result for untested values of  $N$  or  $m$ ; the proof in Section 3 does that.

For  $0 \leq m \leq 7$ , the minor tests interpolate using  $2d + 2$  points, more than enough for the elementary degree bound  $2d$ . They verify the drop to degree  $d$ , both leading coefficients from Lemma 5.1, and additional negative-index values. The recurrence checks include  $0 \leq m \leq 10$  and four nonexceptional parameter pairs, while separate tests examine every exceptional class. The exact two-exponential tests use  $(a, b) = (1, 2)$ ,  $m \leq 7$ , and  $0 \leq N \leq 20$ . These tests check different parts of the derivation, but are not represented as formal proof certificates for arbitrary parameters.

### 10.3 How to reproduce the package

From the archive's top directory, run:

```
python3 code/verify.py --out data
python3 code/catalan_hankel.py 8 3 --a 2 --b 3 --direct
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

Use Python 3.10 or later and do not pass  $-O$ , because the verification suite deliberately uses assertions. The script writes coefficient rows, generating functions, minor polynomials, exact exponential-sector polynomials, and integer sequence tables in JSON or CSV. Rational polynomial coefficients are stored as strings such as  $"8/3"$ , in ascending power order, to avoid any loss of exactness when imported into other systems. The included README documents every file and its indexing convention.

## 11 What has been resolved, and what has not

The all-shift denominator formula is proved here, with a stronger minimality assertion and a complete description of the degenerate parameters. The coefficient expression is likewise proved for all  $N, m$ ; comparison with the exact formula printed in the target source requires the explicit indexing correction in the next appendix. The reciprocal numerator and finite exponential expansions are proved from the same structural determinant calculation.

These are substantive mathematical conclusions about the selected conjectural family. They are not evidence that the Christoffel identity, Jacobi representation, Catalan shifted product, or general rationality are new. A literature search was performed against the source title, arXiv identifier, denominator phrase, and related recurrence work. The cited version still labels the sharp exponent conjectural, and the retrieved

sources did not supply an explicit statement of this full sharp classification. That is a bounded search result, not a proof that no earlier equivalent theorem exists. Before a priority claim or submission, the exact formulas should be checked against the wider orthogonal-polynomial and determinant literature and by a human referee.

Several natural extensions remain separate from the theorem. One may seek a direct combinatorial interpretation of the individual positive integers  $c_{N,m,k}$ , or a product formula for the minors  $M_{m,\ell}(N)$  beyond their first two coefficients. Multiple distinct linear factors in the multiplier lead to more exponential sectors and new collision patterns. A joint limit with growing  $m$  or a parameter approaching  $a = 0$  would require uniform, rather than fixed-parameter, asymptotics. None of these extensions is assumed in the present proofs.

## A The precise correction to the printed conjecture

This appendix distinguishes two kinds of index. The source uses a size- $(n + 1)$  determinant. We write  $N = n + 1$ . To avoid silently identifying incompatible shift conventions, write  $s$  for the source's symbol in its displayed coefficient formula, and reserve  $m$  for our actual Catalan shift.

For  $s \geq 2$  and  $0 \leq k \leq n + 1$ , the coefficient expression printed in Conjecture 2 of [1] can be written as

$$T_{n,k,s} = \frac{C_s}{2s-2} \binom{s+k-2}{s-2} \binom{n+k+2s-2}{2k+2s-3} \times \prod_{j=0}^{\lfloor (s-1)/2 \rfloor - 1} \frac{\binom{2n+2s-2j-1}{2s-4j-5}}{\binom{2s-2j-1}{2s-4j-5}} \prod_{j=0}^{s-3} \frac{2s-j-2}{n+k+2s-j-2}. \quad (\text{A.1})$$

Only the natural, nonambiguous range  $s \geq 2$  is needed here. Empty products have value 1.

**Proposition A.1** (Correct shift). *For  $N \geq 1$ ,  $s \geq 2$ , and  $0 \leq k \leq N$ ,*

$$T_{N-1,k,s} = c_{N,s-1,k}. \quad (\text{A.2})$$

*Consequently, the polynomial using  $T_{N-1,k,s}$  is  $D_N^{(s-1)}(a, b)$ , not  $D_N^{(s)}(a, b)$ . Equivalently, to use that printed expression for an actual shift  $m \geq 1$ , replace its shift symbol by  $s = m + 1$ .*

*Proof.* Taking a ratio of successive terms in (A.1) and cancelling factorials gives

$$\frac{T_{n,k+1,s}}{T_{n,k,s}} = \frac{(n+1-k)(n+k+s+1)}{2(k+1)(2k+2s-1)}. \quad (\text{A.3})$$

This agrees with (2.3) after  $N = n + 1$  and  $m = s - 1$ . It remains to match the initial coefficient. Putting  $n = N - 1$ ,  $k = 0$ , and simplifying the factorials gives

$$T_{N-1,0,s} = C_s \frac{(N+s-1)!}{N!s!} \prod_{j=0}^{p-1} \frac{\binom{2N+2s-2j-3}{2s-4j-5}}{\binom{2s-2j-1}{2s-4j-5}}, \quad p = \left\lfloor \frac{s-1}{2} \right\rfloor. \quad (\text{A.4})$$

At  $N = 1$  this is  $C_s = H_1^{(s)}$ . Its ratio at successive  $N$  is

$$\begin{aligned} \frac{T_{N,0,s}}{T_{N-1,0,s}} &= \frac{N+s}{N+1} \prod_{j=0}^{p-1} \frac{(2N+2s-2j-1)(2N+2s-2j-2)}{(2N+2j+4)(2N+2j+3)} \\ &= \frac{(2N+2s)!(2N+1)!}{(2N+s)!(2N+s+1)!}. \end{aligned} \quad (\text{A.5})$$

In the product, the numerator factors form a consecutive interval of integers, as do the denominator factors; the last equality holds for either parity of  $s$ . The shifted-product formula gives the identical ratio

$$\frac{H_{N+1}^{(s)}}{H_N^{(s)}} = \prod_{i=1}^{s-1} \frac{(2N+2i+1)(2N+2i+2)}{(2N+i+1)(2N+i+2)} = \frac{(2N+2s)!(2N+1)!}{(2N+s)!(2N+s+1)!}.$$

Induction in  $N$  therefore proves  $T_{N-1,0,s} = H_N^{(s)}$ . The ratio (A.3) then proves (A.2) for every  $k$ .  $\square$

The discrepancy is visible in the smallest nontrivial instance. With  $N = 1$  and  $s = 2$ , the printed expression is

$$T_{0,0,2}b + T_{0,1,2}a = 2b + a,$$

whereas the determinant with the stated shift  $s = 2$  is

$$D_1^{(2)}(a, b) = aC_2 + bC_3 = 2a + 5b.$$

At  $a = b = 1$ , these are 3 and 7. The expression  $a + 2b$  is exactly  $D_1^{(1)}(a, b)$ , so the correction is systematic rather than an isolated numerical exception. The original PDF page was inspected as well as its HTML rendering; this is not an artifact of an HTML conversion. The denominator-exponent statement is separate and uses the actual shift  $m$  correctly.

## B Compact reference and archive conventions

| Symbol            | Meaning                                                    |
|-------------------|------------------------------------------------------------|
| $N$               | Determinant size, with $D_0 = 1$                           |
| $m$               | Actual Catalan shift in $aC_{i+j+m} + bC_{i+j+m+1}$        |
| $H_N^{(m)}$       | Pure shifted Catalan determinant                           |
| $d, e, K$         | $\binom{m}{2}$ , $\binom{m}{2} + 1$ , and $(m-1)^2$        |
| $P_m$             | Uniform numerator, not automatically reduced at exceptions |
| $q$               | $1 - (a + 2b)t + b^2t^2$                                   |
| $M_{m,\ell}$      | Minor with column $\ell$ deleted from the binomial block   |
| $Q_m(N; r)$       | Signed generating polynomial of those minors               |
| $r_{\pm}$         | Roots of $r^2 - (a/b + 2)r + 1$                            |
| $\lambda_{\pm}$   | $br_{\pm}$ , the characteristic roots in determinant size  |
| $s$ in Appendix A | Source shift symbol, equal to actual shift plus one        |

The JSON files use ascending polynomial powers. In `coefficient_rows.json`, position  $k$  is the coefficient of  $a^k b^{N-k}$ . In `minor_polynomials.json`, position  $j$  is the coefficient of  $N^j$ . In `two_branch_polynomials.json`, the arrays `P_plus` and `P_minus` describe the polynomial multipliers of  $4^N$  and  $1^N$  for  $(a, b) = (1, 2)$ ; they are not the ordinary-generating-function numerator  $P_m$ .

The source audit records exact URLs, source versions, and search limitations. The tests establish reproducibility of the computation, not priority or formal verification. The archive intentionally contains no third-party article PDFs, no font files, no compiler auxiliaries, and no Python bytecode.

## References

- [1] P. Barry, *Notes on the Hankel transform of linear combinations of consecutive pairs of Catalan numbers*, arXiv:2011.10827v1 (21 November 2020), 29 pp. The denominator conjecture and Conjecture 2 are on p. 4. <https://arxiv.org/abs/2011.10827>.
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- [3] C. Krattenthaler, Hankel determinants of linear combinations of moments of orthogonal polynomials, II, *The Ramanujan Journal* **61** (2023), 597–627; published online 22 November 2021. <https://doi.org/10.1007/s11139-021-00514-8>. See the determinant formula, the confluence discussion, and Section 8 for existing general recurrence theory and its earlier sources.
- [4] NIST Digital Library of Mathematical Functions, Section 18.5, Eq. 18.5.7, Jacobi polynomial hypergeometric representation. <https://dlmf.nist.gov/18.5.E7>.
- [5] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A000108 (Catalan numbers). <https://oeis.org/A000108>.
- [6] OEIS Foundation Inc., *The On-Line Encyclopedia of Integer Sequences*, A001519 and A001906 (Fibonacci subsequences). <https://oeis.org/A001519>; <https://oeis.org/A001906>.

*Online sources checked on 19 September 2026. All theorem numbering in the body refers to the present manuscript, unless a source is explicitly named.*